

Notes: Wang (AER, 2011)

State Misallocation and Housing Prices: Theory and Evidence from China

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1 Big picture

- **Question.** What happens to private-market housing prices when a government privatizes housing that had previously been subsidized and allocated by the state? The paper asks whether the answer depends on *how misallocated* those state housing units were before privatization.
- **Setting.** The empirical setting is China's 1994 urban housing reform, which allowed sitting tenants in state-owned housing to buy the apartments they had been renting from their state employers.
- **Core challenge.** Privatization of state housing does two things at once. It increases private-market *supply* because formerly state-owned units become private assets, but it also changes *demand* because households that had been constrained by state allocation can now move toward their preferred housing bundle. The net price effect is therefore not obvious.
- **Main conceptual contribution.** The paper shows that the key sufficient statistic is prereform *mismatch*: the gap between the housing services households actually consumed under state allocation and the housing services they would have chosen in the private market.
- **Theory prediction.**
 - If state workers were allocated *too little* housing on average, demand shifts out by more than supply after reform, so private-market prices rise.
 - If state allocation was efficient on average, supply and demand shift by the same amount, so prices do not move.
 - If state workers had *too much* housing on average, supply shifts out by more than demand, so prices fall.
- **Main empirical result.** Households in state-owned housing before the reform were consuming housing worth about **15% less** than what they would have chosen in the private market. After the reform, they increased housing consumption, and equilibrium housing prices in the private market rose by about **7.5%**.
- **Behavioral mechanism.** Former state-housing residents responded by moving, renovating, and upgrading their housing quantity and quality. The paper shows that households with larger prereform mismatch adjusted more after the reform.
- **Welfare result.** The paper computes that state-sector households living in allocated housing would have been about **25% better off** if the value of the housing subsidy had been paid as wages and households had been free to choose their housing bundle.
- **Economic intuition.** Subsidized rent alone is not enough to understand the distortion. The real issue is that the state also controlled *who got which unit*. Once households were freed to choose, many wanted more housing than the state had assigned them. That demand reallocation pushed prices up.
- **Takeaway.** Privatizing state property does not have a mechanically negative effect on prices. The sign and size of the effect depend on prereform misallocation.

2 Data and measurement (key objects)

2.1 Why China is a useful laboratory

- Before reform, a large share of urban housing in China was publicly owned and allocated by work units rather than through a market.
- In 1993, about **40%** of urban households in China lived in state-owned housing.
- The 1994 reform was a large, economy-wide privatization that sharply changed property rights and removed subsidized rental distortions.
- During the sample period, direct state allocation mattered far more than mortgage finance. Informal lending was common, but formal mortgage markets were still relatively underdeveloped.

Interpretation. This setting is ideal because the prereform regime featured both distorted prices and nonmarket allocation, while the reform relaxed both simultaneously.

2.2 Institutional background of the reform

Before reform, the state owned all new urban housing stock and allocated units to employees of state-owned enterprises. Rent was highly subsidized, and allocation depended on factors such as job tenure, rank, and social connections.

Beginning in July 1994, state employers were instructed to sell public housing units to sitting tenants. Tenants could buy either full or partial property rights. In the data used in the paper, only about **18%** of households that had been occupying state-owned housing ended up with only partial rights, so most of the reform effectively converted allocated housing into privately controlled assets.

The paper also argues that the reform was largely unanticipated before 1994, drawing on interview evidence and on other empirical work by the author.

2.3 Main data source

The empirical analysis uses the **China Health and Nutrition Survey (CHNS)**.

- Provinces covered: Guangxi, Guizhou, Heilongjiang, Henan, Hubei, Hunan, Jiangsu, Liaoning, and Shandong.
- Survey years: **1989, 1991, 1993, 1997, 2000, 2004**.
- The data therefore contain **three prereform waves** and **three postreform waves**.
- The panel covers roughly **4,400 households**.
- The paper restricts attention to **urban** households because the reform applied to urban housing.

Interpretation. The timing of the panel is unusually convenient: it allows the author to estimate prereform mismatch, then trace household and price responses after the reform.

2.4 How housing services are measured

The paper needs a scalar measure of “housing services” that combines size and quality. It uses the household’s **self-reported market rental value** of its residence as the main proxy.

This measure is attractive because it captures both:

- quantity, such as floor space, and
- quality, such as water access, sanitation, and other amenities.

The rental value is converted into real 1990 RMB.

Interpretation. The paper is not just comparing square meters. It tries to compare the overall value of the housing bundle.

2.5 State housing versus private housing in 1993

Table 1 shows sharp prereform differences:

- Market rental value was lower in state housing: about **89 RMB** per month versus **108 RMB** in private housing.
- Floor space was much smaller in state housing: about **44 square meters** versus **96 square meters** in private housing.
- State units were nevertheless more likely to have modern amenities such as running water and flushing toilets.
- Household income was lower in state housing, but household heads were more educated.

Interpretation. State housing looked like “small but relatively modern” housing. That already hints at nontrivial mismatch: households may have accepted better amenities but too little space.

2.6 The mismatch concept

Let household i ’s allocated state housing service be $\bar{S}_{ci}$ and the amount it would choose in the private market be $d(R_u, X_i)$. Then household-level mismatch in housing services is

$$\delta_i = d(R_u, X_i) - \bar{S}_{ci}.$$

The corresponding value of that mismatch at prereform private-market prices is

$$\Delta_i = R_u \delta_i.$$

- $\Delta_i > 0$: the household was allocated *too little* housing.
- $\Delta_i < 0$: the household was allocated *too much* housing.

The aggregate mismatch for former state-housing residents is $\Delta = \sum_{i=z+1}^q \Delta_i$.

2.7 Dependent variables

The paper studies three kinds of outcomes after measuring prereform mismatch.

1. Residential mobility. Whether a household moved residences or attrited from the survey after the reform.

2. Household housing consumption. Measures of size and quality of the housing bundle, including:

- log floor space,
- flushing toilet,
- drinking water in the home,
- electricity,
- absence of excreta around the dwelling.

3. Market housing prices. The log of market value per square meter for households in private housing.

2.8 A validation variable: postreform mismatch

A crucial internal check is that the author recomputes mismatch *after* reform for households that were living in state-owned housing in 1993. If the key assumption is right – namely that, conditional on observables, state-housing and private-housing households have the same housing preferences – then postreform mismatch for former state-housing residents should be close to zero.

This is exactly what the paper finds: prereform mismatch for state-housing residents is about **0.150**, while postreform mismatch is about **-0.003**, statistically indistinguishable from zero.

Interpretation. This is one of the paper’s strongest credibility checks.

3 Models and empirical specifications (all equations + interpretation)

3.1 Prereform equilibrium

There are two housing markets:

- a controlled market for state-owned housing, indexed c ,
- an uncontrolled private market, indexed u .

Aggregate private-market demand is

$$D_u = \sum_{i=1}^n d(R_u, X_i) + \sum_{i=n+1}^q g(R_u, \bar{R}_c, X_i, \bar{S}_{ci}). \quad (1)$$

Private supply is

$$S_u = f(R_u, F). \quad (2)$$

The prereform private-market equilibrium solves

$$\sum_{i=1}^n d(R_u, X_i) + \sum_{i=n+1}^q g(R_u, \bar{R}_c, X_i, \bar{S}_{ci}) = f(R_u, F). \quad (3)$$

Interpretation. State-housing households affect the private market even before reform because they can always refuse a state unit and enter the private market.

3.2 Postreform equilibrium

After privatization, state-owned housing enters the private market as additional supply:

$$S' = f(R'_u, F) + \bar{S}_c. \quad (4)$$

Postreform aggregate demand becomes

$$D' = \sum_{i=1}^z d(R'_u, X_i) + \sum_{i=z+1}^q d(R'_u, X_i). \quad (5)$$

Hence the new equilibrium price satisfies

$$\sum_{i=1}^z d(R'_u, X_i) + \sum_{i=z+1}^q d(R'_u, X_i) = f(R'_u, F) + \bar{S}_c. \quad (6)$$

Interpretation. The same reform shifts both supply and demand. Whether price rises or falls depends on which shift is larger.

3.3 Why mismatch determines the sign of the price change

Aggregate mismatch at prereform prices is

$$\Delta = \sum_{i=z+1}^q \Delta_i = R_u \sum_{i=z+1}^q \delta_i = R_u \sum_{i=z+1}^q d(R_u, X_i) - R_u \bar{S}_c. \quad (7)$$

The theory then implies three cases:

- If $\Delta > 0$, households in state housing wanted more housing than they were allocated, so demand shifts out by more than supply and $R'_u > R_u$.
- If $\Delta = 0$, the state happened to allocate the efficient bundle on average, so $R'_u = R_u$.
- If $\Delta < 0$, households had more housing than they wanted, so $R'_u < R_u$.

Interpretation. The entire empirical strategy is built around this proposition.

3.4 Estimating mismatch from private-housing demand

The paper estimates the housing demand schedule using only households in private housing in the prereform waves:

$$V_{ijt} = \alpha_0 + \alpha_1 X_{ijt} + \tau_{jt} + \varepsilon_{ijt}, \quad (12)$$

where V_{ijt} is the log monthly market rental value of household i 's housing in province j and year t , X_{ijt} includes:

- a cubic in the age of the household head,
- log household income,
- log total household assets,
- education of the household head,
- province-year indicators.

Using these coefficients, the author predicts how much housing a state-housing household would have chosen in the private market, then defines mismatch as

$$\Delta_{ijt} = \widehat{V}_{ijt} - V_{ijt}.$$

Interpretation. This is a demand-based way to back out how constrained state-housing households were relative to market choice.

3.5 Internal specification with household fixed effects

The paper motivates the validation check with the more general latent-preferences equation:

$$V_{ijt} = \alpha_0 + \alpha_1 X_{ijt} + \tau_{jt} + \gamma_i + \varepsilon_{ijt}, \quad (13)$$

where γ_i captures time-invariant household-specific taste for housing.

If state and private households differed systematically in γ_i , the mismatch estimator would be biased. The paper therefore checks whether postreform mismatch remains different from zero. It does not.

3.6 Mobility regression

To test whether households with larger mismatch were more likely to adjust after the reform, the paper estimates a probit of the form

$$Pr(m_{it} = 1) = g(\beta_1 |\Delta_i| + \beta_2 X_{it} + \tau_t + \varepsilon_{it}), \quad (14)$$

where m_{it} equals one if household i moved or attrited.

Interpretation. Absolute mismatch matters because both “too much housing” and “too little housing” should induce adjustment.

3.7 Household consumption regression

For former state-housing residents, the main quantity-and-quality equation is

$$F_{it} = \alpha_0 + \alpha_1 Post_t \times \Delta_i + \alpha_2 Post_t + \alpha_3 \Delta_i + \alpha_4 X_{it} + \varepsilon_{it}, \quad (15)$$

where F_{it} is either log floor space or one of the quality indicators.

The flexible version is an event-study style specification:

$$F_{it} = \alpha_0 + \sum_{\tau \in \{1991, 1993, 1997, 2000, 2004\}} \beta_\tau (Year_\tau \times \Delta_i) + \alpha_4 X_{it} + \varepsilon_{it}. \quad (16)$$

Interpretation. The postreform interaction should turn positive for households with larger prereform mismatch if reform allows them to realign housing consumption.

3.8 Price regression

To test the equilibrium price prediction, the paper moves to province-level mismatch:

$$R_{ijt} = \alpha_0 + \alpha_1 Post_t \times \Delta_j + \alpha_2 Q_{it} + \tau_t + \gamma_i + \varepsilon_{ijt}, \quad (17)$$

where:

- R_{ijt} is the log value per square meter of household i 's residence,
- Δ_j is average prereform mismatch in province j ,
- Q_{it} controls for changing housing quality,
- γ_i are household fixed effects.

The sample is restricted to households already living in private housing in 1993.

Interpretation. This specification is designed to isolate price changes rather than composition changes in the housing bundle.

3.9 Welfare calculation

Household utility is modeled as Cobb-Douglas:

$$\max_{c, h} (c + \beta x + \varepsilon)^\alpha h^{1-\alpha} \quad \text{such that} \quad y = c + p_h h. \quad (18)$$

The implied housing demand equation is

$$p_h h^* = (1 - \alpha) y^* + \beta(1 - \alpha) x + \varepsilon(1 - \alpha). \quad (19)$$

Using private-housing households to estimate the demand parameters, the paper computes counterfactual welfare if the housing subsidy had been paid as cash wages instead of in-kind allocated housing.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no formal IV here

This paper does *not* use an instrumental-variables design. Credibility comes instead from three sources:

- a sharp institutional reform in 1994,
- a model-based measure of prereform mismatch,
- several internal validation and placebo-style checks.

4.2 Why reverse causality is weak

The core regressor is prereform mismatch, constructed from data before the reform. Postreform changes in mobility, housing consumption, and prices cannot mechanically cause prereform mismatch.

Interpretation. This does not eliminate every concern, but it makes the direction of causality much cleaner than in a simple cross section.

4.3 Why reform timing is plausibly exogenous to household choice

The paper argues that urban residents did not know in advance exactly how or when the state would privatize housing. It cites interview evidence and related work showing that the reform was largely hidden from the public through the early 1990s.

Why this matters. If households had strongly anticipated privatization, prereform housing allocations or mobility decisions might already reflect expected reform. The paper argues against that interpretation.

4.4 The key identifying assumption: conditional preference comparability

The central assumption is that, conditional on observables X , households in state-owned housing and households in private housing have the same housing preferences.

Why it is plausible in this setting:

- many private-housing households were not choosing between state and private housing on the basis of housing taste alone,
- some simply lacked access to state housing because of employment status,
- the comparison is within the same broad urban environment rather than across very different cities or countries.

Why it is credible empirically:

- prereform mismatch for state-housing residents is positive and large,
- postreform mismatch for those same households is essentially zero.

Interpretation. This is the paper’s strongest internal test: once the allocation distortion disappears, the estimated mismatch disappears too.

4.5 Why the price effect is not just province-level omitted trends

A concern is that provinces with higher prereform mismatch might also have had faster income growth or population growth, which independently raised housing prices.

The paper addresses this by adding interactions between time and prereform province GDP and population. The province-level mismatch effect remains similar in magnitude.

Interpretation. The price result is not merely picking up that “fast-growing provinces had both more mismatch and higher prices.”

4.6 Why common quality shifts are not enough to explain the price result

In the price regressions, the dependent variable is value per square meter. Household fixed effects remove time-invariant dwelling quality, and contemporaneous housing-quality controls absorb changing observables such as water, toilet, electricity, and dwelling age.

Interpretation. What remains is intended to be the price component, not simple upgrading of housing characteristics.

4.7 Generated-regressor issue and standard errors

Because mismatch is itself estimated in a first stage, the paper uses **bootstrapped standard errors** in the regressions that include Δ_i or Δ_j .

Interpretation. This is a technically important correction. Without it, uncertainty would be understated.

4.8 Support for the “fixed state supply” assumption

The model assumes the stock of state housing was fixed rather than adjusting to waiting lists. The author reports that province-level data show no correlation between growth of state employment and subsequent construction of state residential housing.

Interpretation. This supports the idea that shortages and misallocation were genuine features of the prereform system, not quickly corrected by new state construction.

4.9 Measurement of self-reported rental values

The appendix checks whether self-reported market rental values are sensible. In hedonic regressions, reported values rise with floor space, running water, flushing toilets, and other quality indicators.

The relationship is similar for private and state-housing households.

Interpretation. The main housing-services variable appears meaningful rather than arbitrary.

4.10 Limitations

The paper is transparent about what it does not capture.

- It does not incorporate labor-market misallocation created by employer-provided housing.
- It abstracts from political or social resources spent obtaining better housing allocations.
- The welfare analysis treats housing primarily as a consumption good, not as a financial asset with capital gains.
- The theory assumes away large moving frictions, though the data suggest adjustment happened partly through renovation as well as mobility.

5 Tables and figures

5.1 Figure 1: urban home ownership over time

Read. Homeownership among urban households rises from roughly **55%** in the early 1990s to above **80%** after the reform.

Economic message. The reform was not marginal. It transformed the structure of urban housing tenure in China.

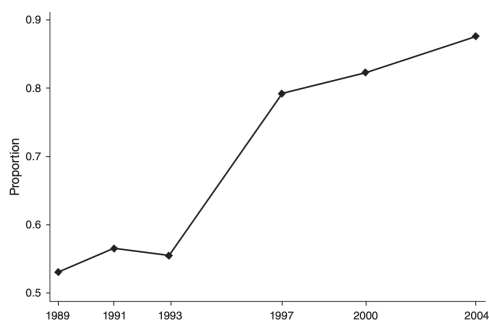


FIGURE 1. URBAN RATES OF HOME OWNERSHIP (Data: CHNS)

TABLE 1—SUMMARY STATISTICS OF URBAN HOUSEHOLDS IN 1993

	In private housing	In state-owned housing
Housing characteristics		
Market rental value	107.75* (99.42)	89.42 (68.22)
Floor space	96.49* (76.61)	44.35 (25.37)
Drinking water	0.78* (0.42)	0.90 (0.29)
Flushing toilet	0.30* (0.46)	0.67 (0.47)
Electricity	0.99 (0.06)	0.99 (0.06)
No excreta around dwelling	0.77* (0.42)	0.94 (0.24)
Water source from plant	0.79 (0.41)	0.84 (0.36)
Age under 20 years	0.80* (0.41)	0.67 (0.47)
Household characteristics		
Household size	3.80* (1.33)	3.34 (1.15)
Durable nonhousing assets	3,729 (5,353)	3,720 (7,875)
Monthly income	624* (1,012)	456 (472)
Age of head	49.06 (14.00)	49.85 (12.83)
Education of head	6.64* (3.98)	9.66 (5.07)
Observations	319	296

Notes: Standard deviations in parentheses, and * denotes the average is significantly different from the other group at the 5 percent level. The data for whether the age of home under 20 years are from the 1991 wave.

5.2 Table 1: descriptive comparison of private and state housing in 1993

Read. State housing had lower rental value and far less floor space, but better amenities. Private housing had much more space, while state housing had more toilets and water access.

Economic message. The state was not merely subsidizing housing – it was allocating a very particular bundle, one that many households would not have chosen on the market.

5.3 Table 2 and Table 3: demand estimation and mismatch measurement

Read.

- The estimated income elasticity of housing demand is about **0.14 to 0.18**.
- Prereform mismatch for state-housing residents is about **0.150**, while private-housing households are normalized to zero.
- Postreform mismatch for former state-housing residents is essentially **zero**.

Economic message. The first-stage demand equation implies that state-housing residents were underconsuming housing relative to market choice before reform, and the validation exercise says this is not just a stable taste difference.

TABLE 2—ESTIMATION OF HOUSING CONSUMPTION AMONG HOUSEHOLDS IN PRIVATE HOUSING

	Prereform (1)	Postreform (2)
Log household income	0.142 [0.035]***	0.183 [0.051]***
Log assets	0.053 [0.011]***	0.138 [0.028]***
Age of head	0.081 [0.059]	−0.032 [0.105]
Age of head ²	−0.001 [0.001]	0.001 [0.002]
Age of head ³	0.000 [0.000]	−0.000 [0.000]
Education of head	0.003 [0.007]	−0.018 [0.010]
Observations	1,212	535
Adjusted R^2	0.17	0.35

Notes: The dependent variable is the logarithm of the market rental value of the household's home. Regressions also include province-year indicators and a constant term. Standard errors clustered by household in brackets.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

TABLE 3—AVERAGE COSTS OF MISMATCH (Δ)

	Prereform	Postreform
State-housing residents	0.150** [0.854]	−0.003 [0.713]
Private-housing residents	0.000 [0.776]	0.000 [0.694]

Notes: Standard deviations in brackets, and ** denotes that the within-column difference with private housing residents is significant at the 1 percent level.

TABLE 4—PROBIT ESTIMATES OF MOVING OR ATTRITING

	Prereform (1)	Postreform (2)
$ \Delta_i $	0.030 [0.018]	0.092 [0.024]***
Age of head	−0.007 [0.005]	−0.021 [0.007]***
Age of head ²	0.000 [0.000]	0.000 [0.000]***
Education of head	−0.003 [0.003]	0.011 [0.003]***
Year 1991	0.020 [0.016]	
Year 2000		−0.046 [0.028]
Year 2004		−0.086 [0.027]***
Observations	1,080	831

Notes: Average marginal effects shown. Bootstrapped standard errors clustered by household in brackets. Regressions also include province fixed effects. The sample in column 1 is limited to households in state-owned housing in the previous period. The sample in column 2 is limited to households in state-owned housing in 1993.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

5.4 Table 4: does mismatch predict moving?

Read. In the postreform period, the marginal effect of absolute mismatch on moving-or-attrition is positive and significant (about **0.092**). In the prereform period the effect is smaller and not statistically significant.

Economic message. Once the allocation distortion is relaxed, households with worse prereform mismatch are more likely to adjust. Before reform, the same households cannot easily respond.

5.5 Table 5: does mismatch predict housing adjustment?

Read.

- A doubling of prereform mismatch is associated with about a **21%** increase in floor space after reform.
- It is also associated with roughly a **14%** higher probability of having a flushing toilet and a **6%** higher probability of having drinking water in the home.
- There is essentially **no effect on electricity**, which makes sense because electricity was already almost universal.
- The flexible event-study style interactions are small before reform and turn positive after reform.

Economic message. The demand side of the mechanism is clearly visible in the data: higher mismatch households move toward larger and better housing bundles once they are free to do so.

5.6 Table 6: province-level mismatch and private-market prices

Read.

- The coefficient on $Post \times \Delta_j$ is about **0.551** in the baseline specification.
- The paper interprets this as saying that a doubling of prereform provincial mismatch raises the postreform price per square meter of housing by about **55%**.
- In the flexible specification, mismatch has essentially no effect in 1991 and 1993, but the effect appears from 1997 onward and remains large through 2004.
- The result survives controls for province-level GDP and population trends.

Economic message. This is the paper's key equilibrium test. Provinces that began with more severe housing misallocation experienced larger postreform price appreciation.

5.7 Table 7 and Table 8: welfare calculation

Read.

- The utility parameter estimate implies $\alpha \approx 0.946$.
- Average prereform welfare is **193.1** for state-housing residents versus **214.4** for private-housing residents.

TABLE 5—IMPACT OF HOUSEHOLD-LEVEL MISMATCH ON HOUSING SIZE AND QUALITY

	Log floor space (1)	Flushing toilet (2)	Drinking water (3)	Electricity (4)	No excreta (5)
<i>Panel A. Parsimonious specification</i>					
Post $\times \Delta_i$	0.21 [0.05]***	0.14 [0.03]***	0.06 [0.02]**	-0.00 [0.00]	0.05 [0.02]***
Post	0.43 [0.13]***	0.13 [0.05]**	-0.02 [0.02]	-0.00 [0.00]	0.16 [0.04]***
Δ_i	-0.19 [0.03]***	-0.20 [0.03]***	-0.07 [0.02]***	0.00 [0.00]	-0.03 [0.02]**
Year 1991	-0.09 [0.07]	-0.10 [0.03]***	-0.03 [0.02]	0.00 [0.00]	0.03 [0.03]
Year 1993	-0.11 [0.07]	0.00 [0.05]	-0.03 [0.02]*	0.00 [0.00]	0.09 [0.04]**
Year 1997	-0.55 [0.35]	0.19 [0.13]	0.27 [0.10]***	0.00 [0.00]	0.16 [0.10]
Year 2000	-0.06 [0.12]	0.00 [0.00]	0.00 [0.00]	-0.00 [0.00]	0.00 [0.01]
Year 2004	0.00 [0.13]	-0.03 [0.04]	0.00 [0.01]	0.00 [0.00]	0.00 [0.00]
Observations	1,867	1,919	1,918	1,914	1,903
Adjusted R^2	0.25	0.34	0.35	0.02	0.05
<i>Panel B. Flexible specification</i>					
$\Delta_i \times 1991$	0.05 [0.04]	-0.03 [0.03]	0.05 [0.04]	0.00 [0.00]	0.01 [0.03]
$\Delta_i \times 1993$	0.10 [0.06]	0.09 [0.04]**	0.03 [0.03]	0.00 [0.00]	-0.01 [0.03]
$\Delta_i \times 1997$	0.26 [0.06]***	0.15 [0.04]***	0.07 [0.04]*	0.00 [0.00]	0.06 [0.03]**
$\Delta_i \times 2000$	0.20 [0.07]***	0.18 [0.05]***	0.12 [0.04]***	0.00 [0.00]	0.03 [0.02]
$\Delta_i \times 2004$	0.33 [0.07]***	0.14 [0.05]***	0.05 [0.04]	-0.00 [0.01]	0.05 [0.02]**
Δ_i	-0.23 [0.04]***	-0.21 [0.04]***	-0.10 [0.03]***	0.00 [0.00]	-0.03 [0.02]
Year 1991	-0.07 [0.08]	-0.12 [0.04]***	-0.01 [0.03]	0.00 [0.00]	0.03 [0.03]
Year 1993	-0.07 [0.08]	0.04 [0.05]	-0.03 [0.02]	0.00 [0.00]	0.08 [0.04]*
Year 1997	-0.17 [0.34]	0.31 [0.12]**	0.24 [0.09]**	-0.00 [0.00]	0.32 [0.092]***
Year 2000	0.37 [0.12]***	0.14 [0.05]**	-0.00 [0.02]	0.00 [0.00]	0.15 [0.04]***
Year 2004	0.46 [0.12]***	0.10 [0.07]	-0.02 [0.02]	-0.00 [0.00]	0.16 [0.04]***
Observations	1,867	1,919	1,918	1,914	1,903
Adjusted R^2	0.26	0.34	0.35	0.02	0.05

Notes: Bootstrapped standard errors clustered by household in brackets. Regressions also include log household income, log assets, a cubic in the head's age, the head's education, province-year indicators, and a constant term. The sample is limited to households that were living in state-owned housing in 1993.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

TABLE 6—IMPACT OF PROVINCE-LEVEL MISMATCH ON THE MARKET PRICE OF HOUSING

	(1)	(2)	(3)	(4)
Post $\times \Delta_j$	0.551 [0.265]**		0.500 [0.208]**	
Year 1991 $\times \Delta_j$		0.125 [0.200]		0.000 [0.000]
Year 1993 $\times \Delta_j$		0.108 [0.493]		0.017 [0.283]
Year 1997 $\times \Delta_j$		0.509 [0.471]		0.187 [1.166]
Year 2000 $\times \Delta_j$		0.658 [0.211]***		0.464 [1.120]
Year 2004 $\times \Delta_j$		0.755 [0.312]*		0.469 [0.463]
Year 1991	0.420 [0.104]***	0.373 [0.085]***	0.421 [0.115]***	2.409 [4.050]
Year 1993	-0.411 [0.246]	-0.458 [0.229]**	-0.432 [0.234]*	-4.946 [14.124]
Year 1997	-0.633 [0.128]***	-0.609 [0.262]**	2.307 [9.067]	4.89 [10.899]
Year 2000	-0.692 [0.152]***	-0.743 [0.160]***	2.254 [9.207]	2.358 [6.911]
Year 2004	-1.193 [0.149]***	-1.275 [0.181]***	1.739 [9.264]	-1.569 [10.266]
Post $\times \log GDP_j$			0.365 [1.275]	
Post $\times \log Population_j$			-0.639 [1.825]	
I(year) $\times \log GDP_j$	No	No	No	Yes
I(year) $\times \log Population_j$	No	No	No	Yes
Observations	2,130	2,130	2,130	2,130
Adjusted R^2	0.44	0.44	0.44	0.45

Notes: The dependent variable is the logarithm of the market value of the home divided by the floor space. Bootstrapped standard errors clustered by province in brackets. Regressions also include the age of home, the source of water, indicator variables for whether the home had running water, a flushing toilet, electricity, excreta around the property, household fixed effects, and a constant term. The sample is limited to households living in private housing in 1993. The fourth column includes the interactions between an indicator for years 1991, 1993, 1997, 2000, 2004, and the logarithm of province-level GDP as well as the logarithm of province-level population.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

- In the counterfactual where housing benefits are paid as wages and households choose freely, state-housing welfare rises to **241.5**.

Economic message. Misallocation was not just a pricing distortion; it materially reduced household welfare.

TABLE 7—ESTIMATION OF THE MONTHLY MARKET RENTAL VALUE OF HOUSING

Monthly household income	0.054 [0.009]***
Total assets	0.001 [0.000]***
Age of head	0.889 [3.037]
Age of head ²	0.023 [0.057]
Age of head ³	-0.000 [0.000]
Education of head	0.505 [0.463]
Observations	4,507
Adjusted R^2	0.12

Notes: Robust standard deviations in brackets clustered at the household level. The regression also includes province-year fixed effects and a constant term. The sample is limited to households living in private housing.

***Significant at the 1 percent level.

TABLE 8—AVERAGE WELFARE OF HOUSEHOLDS

	Observed		Counterfactual
	Prereform	Postreform	
State housing	193.1	290.0	241.5
Private housing	214.4	273.7	
All residents	204.0	281.2	

APPENDIX TABLE A.1—DETERMINANTS OF SELF-REPORTED MARKET RENTAL VALUE (log)

	Pooled sample (1)	Private housing (2)	State controlled (3)
Log floor space	0.374 [0.021]***	0.310 [0.023]***	0.265 [0.048]***
Age 2–4 years	-0.070 [0.062]	0.004 [0.060]	-0.298 [0.158]*
Age 5–9 years	-0.151 [0.064]**	-0.100 [0.062]	-0.303 [0.175]*
Age 10–19 years	-0.130 [0.066]**	-0.113 [0.062]*	-0.254 [0.188]
Age 20+ years	-0.126 [0.061]**	-0.062 [0.058]	-0.317 [0.193]
Electricity	0.265 [0.256]	0.352 [0.387]	0.084 [0.139]
No excreta around home	0.019 [0.036]	0.060 [0.040]	0.077 [0.087]
Running water	0.119 [0.034]***	0.089 [0.039]**	0.195 [0.068]***
Flushing toilet	0.459 [0.028]***	0.468 [0.035]***	0.512 [0.056]***
Water source: Well	-0.029 [0.082]	-0.022 [0.092]	-0.058 [0.195]
Water source: Spring	0.366 [0.092]***	0.497 [0.100]***	0.023 [0.202]
Water source: Plant	0.198 [0.031]***	0.244 [0.036]***	0.197 [0.059]***
Water source: Other	-0.386 [0.198]*	-0.363 [0.299]	-0.574 [0.165]***
R^2	0.26	0.24	0.30
Observations	5,423	3,987	1,429

Notes: Robust standard errors clustered by household in brackets. The regressions also include year and province indicators and a constant term. The omitted category for water source is groundwater.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

5.8 Appendix Table A.1: are self-reported housing values believable?

Read. Log reported rental value rises strongly with floor space, running water, and flushing toilets in both private and state housing. The coefficients look sensible and broadly similar across groups.

Economic message. The main outcome variable – self-reported market rental value – appears informative rather than arbitrary.

6 Short summary

The paper asks how privatizing state-owned housing affects private-market housing prices. The answer is not mechanically negative because privatization changes both housing supply and housing demand. The paper’s key theoretical insight is that the sign of the price effect depends on prereform housing misallocation. If state-housing residents had too little housing relative to what they would have chosen in the market, privatization pushes demand out by more than supply and prices rise.

Using CHNS panel data around China’s 1994 housing reform, the paper estimates that households in state-owned housing were underconsuming housing by about 15% before reform. After privatization, households with greater mismatch were more likely to move and, more importantly, they increased housing consumption and housing quality. Provinces with more prereform mismatch experienced larger increases in private-market housing prices. The estimated aggregate equilibrium effect is a 7.5% increase in housing prices. A counterfactual welfare calculation suggests that allocated housing reduced the welfare of state-housing residents substantially relative to cash wages and free household choice.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes a direct link between prereform housing misallocation and postprivatization housing prices. In China’s case, privatization raised private-market prices because state-housing residents had been allocated less housing than they actually wanted.

7.2 Economic intuition

The common intuition that privatization simply adds supply is incomplete. It is true that former state units become part of the private market. But the reform also frees households from a distorted allocation system. If those households were previously rationed into housing bundles smaller than their preferred bundles, then they enter the private market as stronger demanders of housing. In China, that demand response dominated.

7.3 Takeaway

This is a clean example of why the *allocation rule* matters as much as the *price subsidy*. State intervention can distort not only what people pay, but also what they consume. Once those allocation distortions are removed, equilibrium prices can move in surprising directions.